Problem Set 11

Foundations of Machine Learning

- **Submission**: Turn in as a `PDF` and a **separate source code file** on `http://mycourses.unh.edu`. Scanned handwritten notes are encouraged.
- **Questions**: See the class website for a link to office hours

Relevant reading: MML 6.1–6.5. Problems marked [**code**] should be submitted as a program in a language of your choice (Julia or Python recommended). The cited exercise numbers are from MML (Deisenroth et al., 2021).

1. MML, Chapter 6 exercises: any three on random variables, expectation, and Bayes' theorem (e.g. Exercises 6.1–6.3).
2. MML, Chapter 6 exercises: any two on joint/marginal distributions and the Gaussian (e.g. Exercises 6.4–6.6).
3. [**code**] Draw samples from a two-dimensional Gaussian with a chosen mean and covariance. Estimate the sample mean and covariance from the draws, and produce a scatter plot with the estimated mean marked.
4. **Time spent.** Report approximately how many hours you spent completing this problem set (including the reading).